Two-Sample t-tests, Pt. 1

CSU, Chico Math 351

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outline

outline

Recap: Inference

Statistical inference revolves around a common theme: assume Y is a random variable such that $E(Y) = \mu$ and $Var(Y) = \sigma^2$. We estimate and/or make statements about μ with

$$\bar{Y} \pm t^*_{df} \cdot s_{\bar{Y}}, \quad and \qquad t_{df} = \frac{\bar{Y} - \mu_0}{s_{\bar{Y}}}.$$

outline

Paired Data, definition

Paired data are intimately connected in some way.

Paired Data

Two sets of observations are paired if each observation in one set has a special correspondence or connection with exactly one observation in the other data set.

In other words, we have two random variables, X and Y, but we are specifically testing the difference of each pair of values: $X_1 - Y_1, ..., X_n - Y_n$ as its own, combined sample. The two samples are **paired**.

Paired Data, definition (cont.)

Paired samples generally arise in one of two ways. The first possibility is that, once the i^{th} member of the first sample is chosen, we deliberately choose someone similar (e.g., of the same gender and approximately the same age) to be the i^{th} member of the second sample. This strategy is known as **matching**.

The second possibility is that there is really only one group of subjects, but we obtain measurements for each subject under two different experimental conditions. In either case, we may be interested in testing $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$, where $\mu_1 = E[X_1]$ is the mean for the first population and $\mu_2 = E[X_2]$ is the mean for the second population.

Paired Data, t-test

If we wish to test the hypotheses $H_0 : \mu_1 = \mu_2$ and $H_1 : \mu_1 \neq \mu_2$, we can define μ_{diff} as:

$$\mu_{diff} = E[X_1 - Y_1] = E[X_1] - E[Y_1] = \mu_1 - \mu_2$$

Therefore, $H_0 : \mu_{diff} = 0$ and $H_1 : \mu_{diff} \neq 0$ is equivalent to the hypotheses above.

If the data are paired, their difference forms its own sample X_{diff} of size n; in mathematical terms, $X_{diff} = (X_1 - Y_1, ...X_n - Y_n)$. Therefore we can define our new sample mean and sample error as:

$$\bar{X}_{diff} \quad \text{and} \quad s_{\bar{X}_{diff}}$$

Paired Data, t-test

With a large enough sample size n, we can calculate a test statistic (t-score) on the values in the sample for X_{diff}.

If the sample size is small and X_{diff}'s values are normally distributed, we can instead calculate a t-statistic, performing a **paired t-test**.

$$t = \frac{\bar{X}_{diff} - 0}{s_{\bar{X}_{diff}}/\sqrt{n}}$$

As in the past, we reject H_0 when $2 * (1 - pt(abs(t), n-1)) < \alpha$.

Paired Data, one sample t-test example

Are textbooks actually cheaper online? Compare the price of textbooks at the University of California, Los Angeles' (UCLA's) bookstore and prices at Amazon.com. Seventy-three UCLA courses were randomly sampled in Spring 2010.

```
books <- read.csv("https://raw.githubusercontent.com/
njlytal/MATH351/refs/heads/main/books.csv")
## look at data in RStudio
## what plot should we make?
```

Paired Data, one sample t-test example

Step 1: Plot the data!

```
suppressMessages(library(ggplot2))
ggplot(data = books,  aes(uclaNew - amazNew)) +
geom_boxplot()
or
ggplot(data=books, aes(uclaNew, amazNew)) +
 geom_point() +geom_abline(intercept=0, slope=1)
```

Here, we can obtain a boxplot of the differences in each book's price, or a scatterplot where each dot represents a book and its two prices on the two axes.

Paired Data, one sample t-test example

Calculate and interpret a 95% confidence interval of the difference in Amazon.com versus UCLA's book prices.

Paired Data, one sample t-test example

```
## Define variables
diff <- books$uclaNew-books$amazNew # differences
Xbar <- mean(diff) # sample statistic
n <- length(diff) # sample size
df <- n-1 # degrees of freedom

# Define margin of error
t.star <- qt(1-alpha/2,n-1) # critical value
se <- sd(diff)/sqrt(n) # sample error
ME <- t.star*se # margin of error

## Define confidence interval
CI <- c(Xbar - ME, Xbar + ME)
CI

## [1]  9.435636 16.087652
```

Paired Data, one sample t-test example

We are 95% confident that UCLA's new book prices are greater than Amazon.com's books by between \$9.44 and \$16.09.

Paired Data, one sample t-test example

Set up, evaluate, and conclude in context a hypothesis test at $\alpha = 0.05$.

Paired Data, one sample t-test example

The natural hypotheses are

$$H_0 : \mu_{diff} = 0 \text{ versus } H_1 : \mu_{diff} \neq 0.$$

```
t <- (mean(diff) - 0) / (sd(diff)/sqrt(n)) # test stat
2*(1-pt(abs(t), n-1)) # p-value

## [1] 6.92757e-11
```

Paired Data, one sample t-test example

Because p-value $< 0.0001 < \alpha = 0.05$, we reject H_0. There is insufficient evidence to claim that Amazon.com and UCLA's book prices are the same.

outline

Two Sample t-test

Say we have a sample of size n_1 from one population, and a sample of size n_2 from a second population. Let $X_1, ..., X_{n_1}$ be random variables representing n_1 entries from the first group, and $Y_1, ..., Y_{n_2}$ be random variables representing n_2 entries from the second group. We aren't matching specific entries from each group with each other, so the two samples are **independent**.

Two sample t-tests estimate the difference between two population means from two independent samples of data. We estimate $\mu_1 - \mu_2$ with the point estimator $\bar{X} - \bar{Y}$.

Note that here, we are taking the difference of the overall means, rather than the difference for each pair of observations.

Two Sample t-test Hypotheses

If we let $\mu_1 = E[X_1]$ and $\mu_2 = E[Y_1]$, we can test the hypotheses

$$H_0 : \mu_1 = \mu_2; H_1 : \mu_1 \neq \mu_2$$

or equivalently:

$$H_0 : \mu_1 - \mu_2 = 0; H_1 : \mu_1 - \mu_2 \neq 0$$

Two Sample t-test, standard error

Suppose both populations are normal, but with unequal variances σ_1 and σ_2. We estimate the variance of the point estimator $\bar{X} - \bar{Y}$, namely $Var(\bar{X} - \bar{Y}) = \sigma^2_{\bar{X}-\bar{Y}}$, with

$$s^2_{\bar{X}-\bar{Y}} = \frac{s_x^2}{n_1} + \frac{s_y^2}{n_2}.$$

Two Sample t-test, test statistics

Knowing that

$$s^2_{\bar{X}-\bar{Y}} = \frac{s_x^2}{n_1} + \frac{s_y^2}{n_2}.$$

we can find the test statistic t_{df} as follows:

$$t_{df} = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{s^2_{\bar{X}-\bar{Y}}}}$$

Two Sample t-test, confidence interval

$$(\bar{X} - \bar{Y}) \pm t^*_{df,1-\alpha/2}\sqrt{s^2_{\bar{X}-\bar{Y}}}$$

Two Sample t-test, degrees of freedom

The degrees of freedom will be related to both sample sizes and variances:

$$df = \frac{(s_x^2/n_1 + s_y^2/n_2)^2}{(s_x^2/n_1)^2/(n_1 - 1) + (s_y^2/n_2)^2/(n_2 - 1)}$$

Once more, we reject H_0 when $2 * (1 - pt(abs(t), df)) < \alpha$.

Two Sample t-test, example

Consider the data set `ape::carnivora`. Calculate a 98% confidence interval for the difference in mean longevity between the two superfamilies[1] Caniformia and Feliformia.

```
suppressMessages(library(ape))
data(carnivora)
```

[1] Technically, these are suborders, but they are listed in this data set as superfamilies

Tangent on NAs and R functions

Some helpful functions to help you avoid NAs in R are

```
anyNA()
is.na()
na.omit()
```

Additionally, the dplyr package has the "pipe operator" ($\% > \%$) which lets you chain two functions together. Whatever is to the left of $\% > \%$ will be input as the **first** argument of the function on the right.

```
# UCLA book prices are put as 1st argument of mean()
books$uclaNew %>% mean()

[1] 72.22192
```

Two Sample t-test, example

A 98% CI, difference in longevity between Caniformia and Feliformia.

```
(d <- carnivora %>%
    select(SuperFamily, LY) %>% # only two columns
    na.omit() %>% # be careful with placement
    group_by(SuperFamily) %>% # mean/var by SuperFamily
    summarise(n=n(), v=var(LY), xbar=mean(LY)))

## # A tibble: 2 x 4
##   SuperFamily     n     v  xbar
##   <fct>       <int> <dbl> <dbl>
## 1 Caniformia     24 7139.  192.
## 2 Feliformia     25 2576.  172.
```

This "chain" extracts the variables of interest, isolating the data down to the columns we want and summarizing them for each group.

Two Sample t-test, example

A 98% CI, difference in longevity by Caniformia and Feliformia.

```
alpha <- 0.02
xbarC <- 192.4583; xbarF <- 171.96
diff <- (xbarC - xbarF)  # difference of means
nC <- 24; nF <- 25
vC <- 7139.216; vF <- 2576.04
```

These variables are defined from the output of the previous slide's code.

Two Sample t-test, example

After calculating the degrees of freedom, we can finish the calculations.

```
df <- (vC/nC +vF/nF)^2 /
 ((vC/nC)^2/(nC-1) + (vF/nF)^2/(nF-1))
t.star <- qt(1-alpha/2, df)
se <- sqrt(vC/nC + vF/nF)
ME <- t.star*se

c(diff - ME, diff + ME)
## [1] -28.13848  69.13508
```

Two Sample t-test, example

We are 98% confident that the population difference in mean longevity between the superfamilies Caniformia and Feliformia is between -28.14 and 769.14 months.

Two Sample t-test, example

Set up, evaluate, and conclude in context a hypothesis test at $\alpha = 0.02$.

Two Sample t-test, example

The natural hypotheses are

$$H_0 : \mu_C = \mu_F \text{ versus } H_1 : \mu_C \neq \mu_F.$$

```
# test statistic
t <- ((xbarC - xbarF) - 0) / sqrt(vC/nC + vF/nF)
2*(1-pt(abs(t), df)) # p-value

## [1] 0.3122915
```

NOTE: You can equivalently say $H_0 : \mu_C - \mu_F = 0$

Two Sample t-test, example

Because p-value $= 0.312 > \alpha = 0.02$, we fail to reject H_0. There is insufficient evidence to claim that the true difference in mean longevity between Caniformia and Feliformia is different.

outline

Take Away

Whatever form the problem takes, we're doing the same steps: find parameters of interest, insert point estimates, and calculate the standard error of the estimators.

- Paired data
  - Two variables are intimately connected $\Rightarrow$ their difference has meaning
  - create one variable from the two $\Rightarrow$ one sample t-test
- Two sample data
  - Two variables are independent
  - Point estimate is difference of means
  - Standard error follows from independence
- CLT via standardization makes this all possible.

outline

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Adapted from notes by Dr. Roualdes.